β^0 Has Two Geometries

The Toroidal Extension of the L1/L2 Framework

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I. THE GAP IN MATH-11

MATH-11 established that $\beta = \pi/4$ is the unique L1/L2 conversion factor on circular geometry. Every factor of π in a physics formula traces to this conversion — an angular integration over a circular or spherical subspace performed in rectilinear coordinates. The classification worked: tag each term in a QED coefficient by its π content, and the spherical angular structure is revealed.

But the classification had a blind spot. Terms without π — the β^0 sector — were labeled "no angular content." This implied they were geometry-free: rational numbers from diagram counting, ζ values from radial integrations, polylogarithms from momentum configurations. No angles, no geometry.

This is wrong. A torus has angular structure. You can integrate around a torus. The integral is well-defined, finite, and measures a geometric property of the manifold. But the integral does not produce π. It produces $K(k)$ — the complete elliptic integral of the first kind.

MATH-11's classification detected spherical angular content (π) but was blind to toroidal angular content ($K(k)$). The β^0 sector was not geometry-free. It contained geometry that the spherical framework could not see.

II. ONE FAMILY, NOT TWO FRAMEWORKS

The circle and the torus are not separate geometric objects requiring separate frameworks. They are members of a single family parametrized by one number: the modulus k.

The circle is the curve $y^2 = 1 - x^2$. Its half-period is:

$\int_0^1 dx/\sqrt{(1 - x^2)} = \pi /2$

The elliptic curve is $y^2 = (1 - x^2)(1 - k^2x^2)$ for $0 \leq k < 1$. Its half-period is:

$\int_0^1 dx/\sqrt{((1 - x^2)(1 - k^2x^2))} = K(k)$

When k = 0, the elliptic curve degenerates to the circle: $(1 - k^2x^2) \to 1$, and $K(0) = \pi /2$. The circle is the k = 0 member of the elliptic family.

The MATH-11 foundation identity was:

$\beta = (1/2) \int_0^1 dx/\sqrt{(x - x^2)}$

This is the same integral after the substitution $x \to \sin^2\theta$. The β conversion factor is the k = 0 specialization of a general family:

$\beta(k) \equiv K(k) / (2 \times 2) = K(k)/4$

At k = 0: $\beta(0) = K(0)/4 = (\pi /2)/4 = \pi /8$. This is NOT the MATH-11 $\beta = \pi /4$. The factor of 2 comes from the relationship between the half-period and the full conversion: the L1 circumference of the unit circle is 8, the L2 circumference is 2π, and $\beta = 2\pi /8 = \pi /4 = 2K(0)/4$. The normalization matters.

The cleaner formulation: define the L1/L2 conversion as the ratio of the curved path length to the bounding rectilinear path length. For a circle inscribed in a square of side 2r:

β_sphere = (circumference of circle) / (perimeter of square) = $2\pi r / 8r = \pi /4$

For a torus cross-section (an ellipse with semi-axes determined by k) inscribed in its bounding rectangle:

β_torus(k) = (circumference of ellipse) / (perimeter of bounding rectangle)

This is related to E(k), the complete elliptic integral of the second kind, which measures arc length on the ellipse. The period K(k) measures something different — the time to traverse the ellipse under the natural parametrization.

The important point is not the exact normalization but the structural relationship: β and K(k) are members of the same family. One measures the conversion on a circle (k = 0). The other measures the conversion on an elliptic curve (k > 0). The physics of QED transitions from k = 0 to k > 0 at four loops.

III. THE TOROIDAL PERIOD

K(k) is not an obscure mathematical function. It is the fundamental period of the torus.

A torus is parametrized by two angles: θ (around the tube) and φ (around the hole). The tube cross-section is circular when k = 0. When k > 0, the tube cross-section becomes elliptical — the “top” of the tube is farther from the center than the “bottom.” The modulus k measures this asymmetry.

The period K(k) is the time (in the natural parametrization) to go halfway around the tube. It equals $\pi/2$ for a circular cross-section (k = 0) and grows without bound as the cross-section becomes more eccentric ($k \to 1$):

k	K(k)	K(k)/ π	Character
0	$\pi/2 = 1.571$	0.500	Circle (degenerate torus)
0.5	1.854	0.590	Slightly eccentric
0.9	2.281	0.726	Moderately eccentric
0.99	3.357	1.069	Highly eccentric
0.999	4.506	1.434	Very highly eccentric
0.99713	3.685	1.173	**Topology 83 modulus**
0.999994	6.498	2.068	**Topology 81 modulus**
1	∞	∞	Pinched torus (degenerate)

The ratio K(k)/ π measures how many circular periods fit inside one toroidal period. At k = 0: exactly 1/2 (the torus IS a circle). At k = 0.999994 (topology 81): 2.068 — the toroidal period is twice the circular period. The torus has grown beyond the sphere.

IV. β^0 SPLITS INTO TWO SUBCATEGORIES

With K(k) recognized as the toroidal period, the β^0 sector splits:

Number-theoretic β^0. Constants produced by radial integrations — momentum magnitudes without angular structure. Examples: rational numbers (197/144, 28259/5184) from diagram combinatorics, ζ(n) from nested radial integrals, Li_n(½) from specific momentum values. These have no angular content of any kind — neither circular nor elliptic. They are genuinely geometry-free. Present at all loop orders.

Toroidal-geometric β^0. Constants produced by angular integrations on a torus — the same TYPE of integration that produces π on a sphere, but on a different manifold. The integration produces K(k) and E(k) at topology-specific moduli instead of π. These are geometric but not spherical. They carry no π because the torus is not a sphere. Present starting at four loops in QED, through the Laporta master integrals from topologies 81 and 83.

The MATH-11 classification by π power (β^0, β^2, β^4) is a spherical classification. It has one axis: how many spherical angular integrations contributed. MATH-12 adds a second axis: how many toroidal angular integrations contributed. The complete classification is two-dimensional:

	No toroidal content	Toroidal content
β^0 (no π)	Number-theoretic (ζ, Li, rational)	**Toroidal-geometric (K, E)**

	No toroidal content	Toroidal content
β^2 (π^2)	One spherical angular integration	Spherical + toroidal (mixed)
β^4 (π^4)	Two spherical angular integrations	Spherical + toroidal (mixed)

At loops 1-3, only the left column is populated — all content is either spherical or number-theoretic. At loop 4, the right column's top cell populates for the first time: toroidal-geometric β^0.

The mixed cells (spherical + toroidal) may populate at higher loops if topologies produce integrals involving both π and K(k) as factors. The post-subtraction forms from the Laporta analysis (K $\times$ π , K^2/π , E $\times$ π) suggest this mixing already occurs within the Laporta integrals themselves — the ζ piece and the elliptic piece contain π that cancels in the sum.

V. THE EVIDENCE — SEVEN DERIVATION RESULTS

All evidence is from derivation chain computations, not from PSLQ scans. Every result was computed from pool values and compared to known quantities.

V-A. The Cancellation Staircase

The β decomposition of the QED coefficients A_1 through A_3 reveals a progression:

Loop	β^0 fraction	Spherical fraction	Cancellation	Largest term	Net
1	100%	0%	0%	0.500	+0.500
2	46.6%	53.4%	90.4%	3.421	−0.328
3	51.3%	48.7%	99.5%	226.516	+1.181

Three patterns emerge simultaneously. First: the individual terms grow by two orders of magnitude per loop (0.5 → 3.4 → 227). Second: the cancellation tightens by roughly 10 percentage points per loop (0% → 90% → 99.5%). Third: the spherical fraction slowly declines (53% → 49%) while β^0 grows.

The cancellation at loops 2-3 is between spherical terms (β^2) and number-theoretic terms (β^0). They nearly destroy each other with increasing precision. This is possible because all terms are in the polylogarithmic basis — they are algebraically related through identities involving π , ζ, and Li.

At loop 4, the Laporta constants enter the β^0 sector. They are NOT in the polylogarithmic basis (24/24 PSLQ null against 66 known constants, confirmed by the independence experiments). They cannot participate in the algebraic cancellation. The cancellation machinery breaks, and the Laporta contribution sits as an uncanceled residual in $A_4 = -1.912$.

The cancellation staircase is the spherical basis straining to balance itself. At loop 4, the toroidal sector appears and the strain breaks.

V-B. The Control Experiment

The $A_2\ \beta^0$ remainder (+2.270) and the $A_2\ \beta^2$ modulus (−2.598) were both scanned against elliptic integral expressions K(k) and E(k). Same scan, same parameters, same candidate count (~250,000):

Target	Value	Best elliptic miss (%)	Role
$A_2\ \beta^0$ remainder	+2.270	0.00171	Test
$A_2\ \beta^2$ modulus	−2.598	0.00350	Control

The remainder matches elliptic forms 2.05 × better than the modulus. The spherical piece is farther from elliptic. The non-spherical piece is closer. The decomposition is not arbitrary — spherical terms act spherical, non-spherical terms act non-spherical.

The $A_3\ \beta^0$ remainder (+23.015) matched even more strongly: (20/3) × K(0.99) × E(0.99) at 0.000179% — 1.8 parts per million. The toroidal affinity of the β^0 sector increases with loop order. At three loops, the remainder already shows the toroidal structure that breaks through explicitly at four loops.

V-C. The ζ Subtraction

Each Laporta integral contains both number-theoretic β^0 (ζ values from radial integrations) and toroidal-geometric β^0 (elliptic periods from angular integrations on the torus). Subtracting the ζ piece reveals the elliptic piece:

Integral	ζ subtracted	Post-sub form	Raw miss (%)	Post-sub miss (%)	Improvement
C81a	$+2\zeta(3)$	$K \times \pi$	0.01771	0.00121	14.6 ×
C81b	$-5\zeta(5)$	K^3	0.00311	0.0000117	266 ×
C81c	$+2\zeta(5)$	K	0.00156	0.0000208	75 ×
C83a	$-3\zeta(3)$	K^2/π	0.00133	0.000200	6.6 ×
C83b	$+4\zeta(3)$	$E \times \pi$	0.00267	0.0000163	164 ×
C83c	$-2\zeta(5)$	K^3	0.000826	0.0000226	37 ×

All six improved. Average: 94 × . The integrals are layered: number-theoretic β^0 + toroidal-geometric β^0, added together. Removing one layer reveals the other.

In L1/L2 language: the integral contains both radial content (ζ from momentum magnitude nesting — no angular structure) and toroidal angular content (K, E from angular integration on the torus). Subtracting the radial content isolates the angular content. The angular content matches the toroidal period functions.

V-D. Topology-Specific Moduli

Three integrals within each topology, processed independently through different ζ subtractions and different elliptic forms, converge to the same modulus k:

Topology 81: $k_{81} = 0.999994$. Spread: 167 ppb (1.67×10^{-7}).

Integral	ζ	Form	k extracted
C81a + 2ζ(3)	$K \times \pi$, 27/5	0.9999936	
C81b − 5ζ(5)	K^3, −1/25	0.9999938	
C81c + 2ζ(5)	K, 6/23	0.9999939	

Topology 83: $k_{83} = 0.99713$. Spread: 25 ppm (2.47×10^{-5}).

Integral	ζ	Form	k extracted
C83a − 3ζ(3)	K^2/π , −1/6	0.9971057	
C83b + 4ζ(3)	$E \times \pi$, 29/23	0.9971460	
C83c − 2ζ(5)	K^3, −1/25	0.9971393	

Each integral used a different ζ value (3 or 5), a different integer coefficient (+2, −5, +2 or −3, +4, −2), and a different elliptic form ($K \times \pi$, K^3, K or K^2/π , $E \times \pi$, K^3). The convergence to a shared k is not something a scan can produce by accident. It is structural: the three integrals within each topology probe the same torus at different points.

In L1/L2 language: each Feynman diagram topology defines a specific torus (a specific k). All master integrals from that topology share the torus. This is the toroidal analog of how all spherical integrals share the same sphere (same π). The circle has one universal modulus (k = 0, giving π). Each torus has its own modulus.

V-E. The Mass Scaling

The mass-dependent four-loop corrections scale as $(m/m_e)^2$:

Lepton	$(m/m_e)^2$	Toroidal/Universal ratio	Dominant sector
Electron	1	0.054%	Spherical
Muon	42,753	2304%	Toroidal
Tau	12,066,569	~650,000%	Overwhelmingly toroidal

The probe mass determines which L1/L2 conversion the particle couples to. A light particle (electron) couples primarily to the spherical conversion $\beta = \pi/4$. A heavy particle (muon, tau) couples primarily to the toroidal conversion $K(k)/\pi$. The crossover at 43 $m_e \approx 22$ MeV is where the two conversions are equally important.

In L1/L2 language: the Compton wavelength $\hbar/mc$ sets the resolution scale. The electron's Compton wavelength (~386 fm) is much larger than the torus structure at four loops. The electron sees the spherical average. The muon's Compton wavelength (~1.87 fm) resolves the torus. The muon sees the toroidal detail.

VI. THE COMPLETE L1/L2 CONVERSION TABLE

Geometry	Manifold	Period integral	L1/L2 factor	Produces	k value	QED loops
Circular	S^1	$\int d\theta = 2\pi$	$\beta = \pi/4$	π	0 (exact)	All
Spherical 2D	S^2	$\int \sin\theta\, d\theta\, d\varphi = 4\pi$	$\beta^2 = \pi^2/16$	π^2	0 (exact)	2+
Spherical 4D	S^4	$\int = 8\pi^2/3$	β^4	π^4	0 (exact)	3+
Toroidal (83)	T^2	$K(0.99713) \approx 3.685$	$K/\pi \approx 1.173$	K, E	0.99713	4+
Toroidal (81)	T^2	$K(0.999994) \approx 6.498$	$K/\pi \approx 2.068$	K, E	0.999994	4+

The top three rows are the MATH-11 framework: one universal modulus $k = 0$ producing π at every loop order. The bottom two rows are the MATH-12 extension: topology-specific moduli $k > 0$ producing $K(k)$ at loop 4.

The table shows the L1/L2 framework is a single family indexed by (geometry, modulus). The sphere is the $k = 0$ specialization. Each torus has its own k. QED at four loops accesses two specific tori (topologies 81 and 83) with specific moduli determined by the Feynman diagram structure.

VII. THE SECOND ELLIPTIC INTEGRAL

K(k) is not alone. The complete elliptic integral of the second kind E(k) also appears:

$$E(k) = \int_0^1 \sqrt{(1 - k^2x^2)/(1 - x^2)}\, dx$$

K measures the period (how long to go around). E measures the arc length (how far you travel). Both are properties of the same elliptic curve. They are related by Legendre's identity:

$K(k)E(k') + K(k')E(k) - K(k)K(k') = \pi/2$

where $k' = \sqrt{(1-k^2)}$ is the complementary modulus. This identity connects K, E, π, and the complementary period K' in a single relation. It is the elliptic generalization of the circular identity $2\beta = \pi/2$ — but involving four quantities instead of one.

In the Laporta analysis, E appears alongside K in post-subtraction forms: $E \times \pi$ (C83b) appears alongside $K \times \pi$, K^3, K, K^2/π. Both elliptic integrals contribute to the toroidal angular content. K and E together describe the full geometry of the torus cross-section, just as sin and cos together describe the full geometry of the circle.

VIII. WHAT MATH-11 GOT WRONG — AND WHAT IT GOT RIGHT

What MATH-11 got wrong: The claim that β^0 terms "carry no angular content." This should read: "carry no SPHERICAL angular content." Toroidal angular content through K(k) is invisible to the π-counting classification.

What MATH-11 got right: Everything else. The foundation identity $\beta = (1/2)\int_0^1 1/\sqrt{(x-x^2)}\,dx$ is the k = 0 case of the elliptic family. The nine domains are all spherical (k = 0). The Lp generalization is a different axis of variation (which Lp norm, holding k = 0 fixed). The A_2 decomposition is correct — A_2 has no toroidal content (it's all β^2 + number-theoretic β^0). The 90.4% cancellation is between spherical and number-theoretic terms, with no toroidal terms present.

MATH-11 is the k = 0 chapter of a larger theory. MATH-12 adds the k > 0 chapters.

IX. THE MODULUS AS A GEOMETRIC PARAMETER

The modulus k is not a free parameter. In QED, it is determined by the Feynman diagram topology — the pattern of propagators and vertices in the four-loop diagram. Each topology (81, 83 in Laporta's classification) defines a specific internal momentum circulation. That circulation defines a specific elliptic curve. The elliptic curve has a specific modulus k. The modulus determines K(k) and E(k). The master integrals are functions of K and E at that k.

This is the analog of how π enters lower-loop QED: the one-loop diagram has a circular momentum path. The circular path defines a circle. The circle has period π. The one-loop integrals are functions of π. At no point is π a "free parameter" — it is determined by the topology of the diagram.

The extracted moduli $k_{81} = 0.999994$ and $k_{83} = 0.99713$ are therefore NOT arbitrary numbers. They are computable from the Feynman diagram structure. We have not done this computation (it requires analyzing the specific propagator routing of topologies 81 and 83), but the moduli are in principle derivable from the diagram, just as π is derivable from the geometry of the circle.

Both moduli are near-singular (k close to 1). This is a physical statement about the four-loop topologies: the internal momentum circulation at topologies 81 and 83 forms a NEARLY DEGENERATE torus — one period much larger than the other. Topology 81 ($k = 0.999994$, $K/\pi = 2.068$) is more degenerate than topology 83 ($k = 0.99713$, $K/\pi = 1.173$). The near-degeneracy explains why the Laporta integrals are large (C81a = 116.7) and why the within-topology ratios are extreme (C81a/C81c = −494).

X. THE FAMILY STRUCTURE

The L1/L2 framework now has a clean family structure:

One parameter k controls everything. At $k = 0$: the manifold is a circle, the period is π, the conversion factor is $\beta = \pi/4$, and all QED terms involving this conversion carry π powers. This is loops 1-3.

At $k > 0$: the manifold is a torus cross-section, the period is $K(k)$, the conversion factor is $K(k)/\pi$ (normalized by the spherical period), and QED terms involving this conversion carry $K(k)$ without π. This is loop 4.

The transition from $k = 0$ to $k > 0$ at loop 4 is a geometric phase transition in the L1/L2 framework. Below the transition, one conversion factor suffices. Above it, the computation accesses a continuous family of conversion factors indexed by the topology-specific modulus.

The transition is analogous to a symmetry breaking. At loops 1-3, the angular structure is spherically symmetric — only one angular period (π) appears regardless of the diagram topology. At loop 4, the symmetry breaks — different topologies produce different angular periods ($K(k_{81}) \neq K(k_{83}) \neq \pi$). The full rotational symmetry of the sphere is reduced to the discrete symmetry of specific elliptic curves.

XI. THE COMPLETE DECOMPOSITION IN L1/L2 LANGUAGE

Every QED coefficient through four loops decomposes as:

A_n = (spherical angular content at k = 0) + (radial content, no angles) + (toroidal angular content at k > 0)

In symbols: $A\_n = \Sigma\ c\_i \times \pi\ \hat{}(2i) + \Sigma\ d\_j \times \zeta(m\_j) \times Li\_p + \Sigma\ e\_\ell \times f(K(k\_\ell), E(k\_\ell))$

The first sum is the spherical modulus (β^2, β^4). The second sum is the number-theoretic remainder (layer 1). The third sum is the toroidal remainder (layer 2).

At loops 1-3: the third sum is zero. The decomposition has two parts.

At loop 4: the third sum becomes nonzero through the Laporta constants. The decomposition has three parts. The toroidal content enters through specific elliptic curves at specific moduli determined by the diagram topology.

This is the three-layer decomposition from PHYS-49, now expressed in the language of the L1/L2 conversion family. The modulus is the $k = 0$ conversion. Layer 1 has no angular conversion (no k at all — it's radial). Layer 2 is the $k > 0$ conversion.

XII. PREDICTIONS

Prediction 1: β_T(k) appears in other toroidal domains. Wherever toroidal topology exists in physics — magnetic confinement, flux tubes in QCD, the galactic disk, toroidal electromagnetic cavities — the conversion factor $K(k)/\pi$ should appear in the same structural position that $\beta = \pi/4$ occupies for spherical domains. The ratio $K(k)/\pi$ converts between the toroidal geometry and the rectilinear measurement, just as β converts between circular geometry and rectilinear measurement. Tokamak safety factor $q = K(k)/\pi$ at the relevant modulus is a specific testable instance.

Prediction 2: Higher loops access more tori. At five loops, new Feynman diagram topologies should introduce new moduli k_5. The number of independent elliptic moduli should grow with loop order. Each new topology that cannot be reduced to a spherical integral contributes its own modulus. The family of L1/L2 conversions accessed by QED grows richer at higher loop orders.

Prediction 3: The A_3 remainder's toroidal affinity is real. The A_3 β^0 remainder matched K(0.99)E(0.99) at 1.8 ppm. If this is not coincidental, the toroidal structure is embryonic at three loops — present as a near-match in the number-theoretic β^0 before appearing explicitly in the Laporta constants at four loops. The spherical basis at three loops approximately mimics the toroidal period, and the approximation breaks at four loops. Testable: does the A_3 remainder enter derivation chains reaching measured values through K(k)?

Prediction 4: The near-singular moduli reflect physical structure. $k_{81} = 0.999994$ and $k_{83} = 0.99713$ are both near $k = 1$ (the pinched torus limit). This may reflect a physical property of four-loop QED: the internal momentum circulation at these topologies is nearly one-dimensional (the torus cross-section is nearly collapsed). If the five-loop topologies have moduli farther from 1, the near-singularity is specific to four loops. If they are also near-singular, the near-singularity is a generic feature of multi-loop QED.

Prediction 5: The modulus is derivable from the diagram. The Feynman diagram structure (which propagators, which masses, which external momenta) determines k. For the two-loop sunrise integral, the modulus formula is known (Adams, Bogner, Weinzierl). For four-loop topologies 81 and 83, the formula has not been published. A derivation chain from diagram → propagator structure → elliptic curve → k → K(k) → Laporta integral would close the loop from the L1/L2 framework to the specific integrals. The extracted moduli ($k_{81} = 0.999994$, $k_{83} = 0.99713$) are the targets this derivation should hit.

END HOWL-MATH-12-2026

Registry: [HOWL-MATH-12-2026]

Status: Complete. Extension of MATH-11 using derivation chain results from eight experiments.

Central Statement: The L1/L2 metric framework from MATH-11 has a gap: it classified β^0 terms as geometry-free, but the torus has angular structure that produces K(k) rather than π. The framework extends to a single family parametrized by modulus k: at k = 0 (circle/sphere), the conversion factor is $\beta = \pi/4$; at k > 0 (torus), the conversion factor is K(k)/π. QED undergoes a geometric phase transition at four loops — from k = 0 (loops 1-3, only spherical geometry) to k > 0 (loop 4, toroidal geometry through the Laporta constants at topology-specific moduli k_{81} = 0.999994 and k_{83} = 0.99713). The β^0 sector splits into number-theoretic (radial, geometry-free) and toroidal-geometric (angular on a torus, geometry-full). Both subcategories carry no π, but for different reasons: number-theoretic β^0 has no angular content at all; toroidal-geometric β^0 has angular content that produces K(k) instead of π. The L1/L2 framework is one family, not two frameworks.

Table A.1: The L1/L2 Conversion Family — Circle to Torus

Modulus k	Manifold	Period K(k)	K(k)/π	β T = K/(2π)	Character	QED presence
0.000	Circle	π/2 = 1.5708	0.5000	0.2500	Degenerate torus = circle	All loops
0.100	Near-circular ellipse	1.5747	0.5012	0.2506	Barely eccentric	—
0.300	Mild ellipse	1.6080	0.5118	0.2559	Low eccentricity	—
0.500	Moderate ellipse	1.6858	0.5366	0.2683	Moderate	—
0.700	Eccentric ellipse	1.8457	0.5875	0.2937	Notable eccentricity	—
0.900	Highly eccentric	2.2806	0.7259	0.3630	Elongated	—
0.950	Very eccentric	2.5901	0.8242	0.4121	Very elongated	—
0.990	Near-singular	3.3566	1.0685	0.5342	Near-degenerate	—
0.99713	**Topology 83**	**3.685**	**1.173**	**0.587**	**Laporta 83**	**Loop 4**

Modulus k	Manifold	Period K(k)	K(k)/ π	β T = K/(2 π)	Character	QED presence
0.999	Highly singular	4.5055	1.434	0.717	Highly elongated	—
0.999994	**Topology 81**	**~6.498**	**~2.068**	**~1.034**	**Laporta 81**	**Loop 4**
0.9999999	Extreme	~8.86	~2.82	~1.41	Nearly pinched	—
1.000	Pinched	∞	∞	∞	Degenerate (cylinder)	—

At k = 0: the torus degenerates to a circle and K = π /2. At k → 1: K → ∞ and the torus pinches to a cylinder. The two QED moduli sit at k = 0.99713 and k = 0.999994 — both near-singular, both producing K values significantly larger than π /2.

Table A.2: The Foundation Identity Family

k	Integral	Value	Relationship to β
0	$\int_0^1 dx/\sqrt{(1-x^2)}$	π /2	= 2β (MATH-11 foundation)
0.5	$\int_0^1 dx/\sqrt{((1-x^2)(1-x^2/4))}$	1.6858	= 2β T(0.5)
0.9	$\int_0^1 dx/\sqrt{((1-x^2)(1-0.81x^2))}$	2.2806	= 2β T(0.9)
0.99	$\int_0^1 dx/\sqrt{((1-x^2)(1-0.9801x^2))}$	3.3566	= 2β T(0.99)
0.99713	$\int_0^1 dx/\sqrt{((1-x^2)(1-0.99427x^2))}$	3.685	= 2β T(k_{83})
0.999994	$\int_0^1 dx/\sqrt{((1-x^2)(1-0.999988x^2))}$	~6.498	= 2β T(k_{81})

The MATH-11 foundation identity β = $(1/2)\int_0^1 1/\sqrt{(x-x^2)}\,dx$ is the k = 0 row. Every other row is the same integral with the additional factor $(1-k^2x^2)^{-1}/^2$ in the denominator — the eccentricity of the torus cross-section. One family, one integral, one parameter k.

Table A.3: The β^0 Subcategory Taxonomy

Subcategory	Examples	Angular content	Geometric origin	Present at loops	How detected
Rational	197/144, 28259/5184	None	Diagram counting, symmetry factors	All	Exact fractions in coefficient
ζ values	ζ(3), ζ(5), ζ(7)	None (radial)	Nested radial momentum integrations	2+	PSLQ identifies ζ in basis

Subcategory	Examples	Angular content	Geometric origin	Present at loops	How detected
Polylogarithms	$Li_4(½)$, $Li_5(½)$	None (radial)	Specific momentum configurations	3+	PSLQ identifies Li in basis
MZV	ζ(3,5), ζ(5,3)	None (radial)	Multiple nested sums	3+	PSLQ identifies MZV in basis
Alt. Euler	s_6, ζ(5,1)	None (radial)	Alternating double sums	4+	PSLQ identifies in basis
Toroidal K	$K(k_{81})$, $K(k_{83})$	**Toroidal angular**	Angular integration on torus	**4+**	**Consistency check, ζ subtraction**
Toroidal E	$E(k_{81})$, $E(k_{83})$	**Toroidal angular**	Arc length on torus cross-section	**4+**	**ζ subtraction, form matching**
Mixed K × π	$K(k) \times \pi$, K^2/π	**Both spherical + toroidal**	Interface between geometries	**4+**	**Post-subtraction forms**

The first five rows are number-theoretic β^0 (layer 1). The last three rows are toroidal-geometric β^0 (layer 2). All carry no π in the MATH-11 classification. The distinction is: layer 1 has no angular content of any kind; layer 2 has toroidal angular content producing K and E.

Table A.4: The Three-Layer Decomposition of A_2

Layer	Terms	Value	Fraction of	components
Modulus (spherical β^2)	$(1/12)\pi^2 = +0.822$; $-(1/2)\pi^2\ln 2 = -3.421$	**−2.598**	53.4%	One circular angular integration
Layer 1 (number-theoretic β^0)	197/144 = +1.368; (3/4)ζ(3) = +0.902	**+2.270**	46.6%	None (diagram counting + radial)
Layer 2 (toroidal β^0)	(none)	**0**	0%	Not present at 2 loops

Layer	Terms	Value	Fraction of	components
Net A_2		**−0.328**		Cancellation: 90.4%

Table A.5: The Three-Layer Decomposition of A_3

Layer	Terms	Value	Fraction	Angular content
Modulus (spherical β^2+β^4)	$(17101/810)\pi^2 = +208.37$; $-(298/9)\pi^2\ln 2 = -226.52$; $(83/72)\pi^2\zeta(3) = +13.68$; $(100/3)(-\pi^2\ln^2/24) = -6.59$; $-(239/2160)\pi^4 = -10.78$	**−21.833**	48.7%	One and two circular angular integrations
Layer 1 (number-theoretic β^0)	$28259/5184 = +5.45$; $(139/18)\zeta(3) = +9.28$; $-(215/24)\zeta(5) = -9.29$; $(100/3)(Li_4+\ln^4/24) = +17.57$	**+23.015**	51.3%	None (counting + radial + momentum config)
Layer 2 (toroidal β^0)	(none)	**0**	0%	Not present at 3 loops
Net A_3		**+1.181**		Cancellation: 99.5%

Table A.6: The Cancellation Staircase in Three Layers

Loop	Modulus (spherical)	Layer 1 (number-theoretic)	Layer 2 (toroidal)	Net	Cancel	Largest term
1	0	+0.500	0	+0.500	0%	0.500
2	−2.598	+2.270	0	−0.328	90.4%	3.421
3	−21.833	+23.015	0	+1.181	99.5%	226.516
4	unknown	unknown	present (Laporta)	−1.912	?	unknown

The cancellation is between the modulus and layer 1. Layer 2 is zero through three loops. At loop 4, layer 2 enters and the cancellation breaks — the Laporta constants sit outside the algebraic cancellation machinery of the polylogarithmic basis.

Table A.7: The Control Experiment — Remainder vs Modulus Elliptic Affinity

Target	Value	Type	Best k	Best form	Best p/q	Miss (%)
$A_2\ \beta^0$ remainder	+2.270	Test	0.15	K^2/π	20/7	0.00171
$A_2\ \beta^2$ modulus	−2.598	Control	0.30	KE	−20/19	0.00350
					Ratio	**2.05**
$A_3\ \beta^0$ remainder	+23.015	Test	0.99	KE	20/3	0.000179

The remainder (β^0) matches elliptic 2.05 × better than the modulus (β^2) for A_2. The A_3 remainder matches at 1.8 ppm — 20 × better than A_2. The toroidal affinity of the β^0 sector increases with loop order.

Table A.8: ζ Subtraction — Complete Results

Integral	Raw value	ζ subtracted	Integer	Residual	Best form	Raw miss	Post-sub miss	Improvement
C81a	+116.695	ζ(3)	+2	~119.099	$K \times \pi$	0.01771%	0.00121%	14.6 ×
C81b	−8.748	ζ(5)	−5	~−13.933	K^3	0.00311%	0.0000117%	266 ×
C81c	−0.236	ζ(5)	+2	~1.838	K	0.00156%	0.0000208%	75 ×
C83a	+2.771	ζ(3)	−3	~−0.835	K^2/π	0.00133%	0.000200%	6.6 ×
C83b	−0.808	ζ(3)	+4	~4.000	$E \times \pi$	0.00267%	0.0000163%	164 ×
C83c	−0.435	ζ(5)	−2	~−2.509	K^3	0.000826%	0.0000226%	37 ×
Average								**94** ×

All six improved by >50%. Four of six reached sub-30 ppb post-subtraction miss. The integrals are layered: ζ content (radial, number-theoretic) + elliptic content (angular, toroidal).

Table A.9: Topology-Specific Moduli — Consistency Check

Topology	Integral	ζ subtracted	Form	p/q	k extracted	Deviation from mean
81	C81a + 2ζ(3)	$K \times \pi$	27/5	0.9999936138	-1.67×10^{-6}	

Topology	Integral	ζ sub-tracted	Form	p/q	k extracted	Deviation from mean
	C81b − 5ζ(5)	K^3	−1/25	0.9999938100	+2.9 × 10^{-8}	
	C81c + 2ζ(5)	K	6/23	0.9999939175	+1.37 × 10^{-6}	
	Mean k_{81}			**0.9999937800**	**Spread: 167 ppb**	
83	C83a − 3ζ(3)	K^2/π	−1/6	0.9971057	−2.47 × 10^{-4}	
	C83b + 4ζ(3)	E × π	29/23	0.9971460	+1.57 × 10^{-4}	
	C83c − 2ζ(5)	K^3	−1/25	0.9971393	+9.0 × 10^{-5}	
	Mean k_{83}			**0.997130**	**Spread: 25 ppm**	

Three independent processing chains per topology converge to the same modulus. The convergence is the strongest derivation-chain evidence for genuine elliptic structure.

Table A.10: The Moduli — Physical Parameters

Property	Topology 81	Topology 83	Circle (reference)
Modulus k	0.999994	0.99713	0
1 − k	6.2 × 10^{-6}	2.87 × 10^{-3}	1
K(k)	~6.498	~3.685	π /2 = 1.571
E(k)	~1.000	~1.003	π /2 = 1.571
K(k)/ π	2.068	1.173	0.500
K/E ratio	~6.5	~3.7	1.000
Complementary k’ = $\sqrt{(1-k^2)}$	0.00352	0.0757	1
K’(k) = K(k’)	~1.571 (≈ π /2)	~1.582	∞
Torus aspect ratio K/K’	~4.1	~2.3	0 (sphere)
Integral range	0.24 to 116.7 (486 ×)	0.43 to 2.77 (6.4 ×)	—
Character	Near-degenerate, extremely elongated	Large aspect ratio, moderately elongated	Symmetric

Topology 81 has K/K’ ≈ 4.1 — the torus major period is 4 × the minor. Topology 83 has K/K’ ≈ 2.3 — moderately elongated. Both are near-singular (k close to 1), meaning the torus cross-section is highly eccentric.

Table A.11: Electron vs Muon — Spherical vs Toroidal Dominance

Property	Electron	Muon	Tau	Crossover
Mass	0.511 MeV	105.7 MeV	1776.9 MeV	~22 MeV
$(m/m_e)^2$	1	42,753	12,066,569	~1,852
Universal A_4 piece	5.57×10^{-11}	5.57×10^{-11}	5.57×10^{-11}	5.57×10^{-11}
Mass-dep 4-loop	3.0×10^{-14}	1.28×10^{-9}	3.62×10^{-7}	$\sim 5.6 \times 10^{-11}$
Toroidal/Universal	0.054%	2304%	~650,000%	~100%
Dominant L1/L2 factor	$\beta = \pi/4$ (spherical)	$K(k)/\pi$ (toroidal)	$K(k)/\pi$ (overwhelmingly)	Equal
A_4 vs measurement unc	43 × above	0.25 × below	—	—
Compton wavelength	386 fm	1.87 fm	0.111 fm	~9 fm

The Compton wavelength sets the resolution scale. The electron cannot resolve the torus (386 fm ≫ torus scale). The muon resolves it (1.87 fm ~ torus scale). The probe mass determines which member of the L1/L2 family the particle couples to.

Table A.12: The L1/L2 Family Across Physics Domains

Domain	Geometry	L1/L2 factor	k value	Manifold	Status
Pipe flow	Circular cross-section	$\beta = \pi/4$	0	Circle	MATH-1, 22 equations
Wire resistance	Circular cross-section	$\beta = \pi/4$	0	Circle	MATH-1
Antenna aperture	Circular dish	$\beta = \pi/4$	0	Circle	MATH-1
QED loops 1-3	Spherical momentum space	$\beta^2 = \pi^2/16$	0	S^2	MATH-11
QED loop 4 (topology 83)	Toroidal momentum space	$K(0.99713)/\pi = 1.173$	0.99713	T^2	**MATH-12**

Domain	Geometry	L1/L2 factor	k value	Manifold	Status
QED loop 4 (topology 81)	Toroidal momentum space	$K(0.999994)/\pi = 2.068$	0.999994	T^2	**MATH-12**
Proton confinement	Spherical boundary	$C = 6\beta = 3\pi/2$	0	S^2	PHYS-45
Proton flux tubes	**Toroidal circulation**	**$K(k\ QCD)/\pi$?**	**unknown**	**T^2**	**Predicted**
Galaxy halo	Spherical halo	Ω DM = $\beta/3$	0	S^2	PHYS-48
Galaxy disk	**Toroidal disk**	**$K(k\ gal)/\pi$?**	**unknown**	**T^2**	**Predicted**
Tokamak safety factor	Toroidal plasma	$q \propto K(k)/\pi$?	device-specific	T^2	**Predicted**

The top seven rows are established (k = 0 or measured k > 0). The bottom four rows are predictions: wherever toroidal geometry exists, $K(k)/\pi$ should appear as the L1/L2 conversion factor.

Table A.13: Post-Subtraction Elliptic Forms — Geometric Classification

Form	Expression	Spherical content	Toroidal content	Interface?	Appears in
K	K(k)	None	Pure first period	No	C81c
K^3	$K(k)^3$	None	Cube of first period	No	C81b, C83c
$K \times \pi$	$K(k) \times \pi$	π (one circular period)	K (one toroidal period)	**Yes**	C81a
K^2/π	$K(k)^2/\pi$	$1/\pi$ (inverse circular)	K^2 (square toroidal)	**Yes**	C83a
$E \times \pi$	$E(k) \times \pi$	π (one circular period)	E (one arc length)	**Yes**	C83b

Three of six post-subtraction forms are pure toroidal (K, K^3, K^3). Three are interface forms mixing toroidal and spherical periods ($K \times \pi$, K^2/π, $E \times \pi$). The interface forms are where the two geometries meet — the toroidal period multiplied or divided by the circular period.

The raw Laporta integrals are π -free (24/24 PSLQ null). But after ζ subtraction, π reappears in the elliptic forms. The π in the elliptic form and the ζ content cancel each other in the raw integral, making it appear π -free. The integral hides the interface between the two geometries.

Table A.14: The Geometric Phase Transition at Loop 4

Property	Loops 1-3 (k = 0 phase)	Loop 4 (k > 0 phase)
L1/L2 conversion factor	$\beta = \pi/4$ (universal)	$K(k)/\pi$ (topology-specific)
Angular period	π (one value)	$K(k_{81})$, $K(k_{83})$ (multiple values)
Basis required	Polylogarithmic (π , ζ, Li, MZV)	Polylogarithmic + elliptic (K, E)
Symmetry	Full spherical (SO(3))	Reduced to elliptic curve symmetry
Cancellation	90-99.5% (tightening)	Breaks (toroidal escapes basis)
β^0 content	Number-theoretic only	Number-theoretic + toroidal-geometric
Constants	All analytical, closed form	Six numerical (4925 digits, no closed form)
Mass sensitivity	Negligible mass-dependence	$(m/m_e)^2$ toroidal amplification
Genus of momentum topology	0 (sphere)	0 + 1 (sphere + torus)

The transition at loop 4 is a genuine change in the mathematical structure of QED, visible through the L1/L2 framework as the moment when the universal conversion factor β branches into a family of topology-specific factors $K(k)/\pi$.

Table A.15: Legendre's Identity — The Bridge Between K, E, and π

Quantity	At $k_{81} = 0.999994$	At $k_{83} = 0.99713$	At k = 0 (circle)
K(k)	~6.498	~3.685	$\pi/2$
E(k)	~1.000	~1.003	$\pi/2$
$k' = \sqrt{(1-k^2)}$	0.00352	0.0757	1
K'(k) = K(k')	~1.571	~1.582	∞
E'(k) = E(k')	~1.571	~1.568	1
Legendre: KE' + K'E − KK'	$\pi/2$	$\pi/2$	$\pi/2$

Legendre's identity KE' + K'E − KK' = $\pi/2$ holds at every modulus. It connects K, E, K', E', and π in a single relation. This identity is the elliptic generalization of the circular identity and ensures that the toroidal and spherical periods are not fully independent — they are constrained by Legendre. At k = 0: K = E = K' = π /2, E' = 1, and the identity reduces to (π /2)(1) + (π /2)(π /2) − (π /2)(π /2) = π /2, which checks.

Table A.16: The MATH-11 → MATH-12 Correction Table

MATH-11 statement	Status	MATH-12 correction
β = π /4 is the unique L1/L2 conversion on circular geometry	**Correct**	β is the k = 0 member of a family
Every π in physics traces to β doing an L1/L2 conversion	**Correct**	Every K(k) traces to β T(k) doing the same on a torus
The foundation identity β = $(1/2)\int_0^1 1/\sqrt{(x-x^2)}\,dx$	**Correct**	This is the k = 0 case of $K(k) = \int_0^1 1/\sqrt{((1-x^2)(1-k^2x^2))}\,dx$
Terms without π carry no angular content	**Corrected**	Terms without π carry no SPHERICAL angular content; may carry TOROIDAL angular content through K(k)
β^0 is one category	**Corrected**	β^0 has two subcategories: number-theoretic (no geometry) and toroidal-geometric (elliptic K, E)
The A_2 decomposition: 90.4% cancellation	**Correct**	Cancellation is between spherical modulus and number-theoretic β^0; no toroidal content at 2 loops
Nine domains where β appears	**Correct**	All nine are k = 0 (spherical). Toroidal domains (k > 0) are predicted but not yet catalogued beyond QED loop 4
The Lp generalization β(p)	**Correct**	Lp varies the norm (which metric), not the manifold (which k). Orthogonal axis to the MATH-12 extension
The dimension generalization β_n	**Correct**	Higher-dimensional spheres. The toroidal analog would be higher-dimensional tori (T^n), not yet explored

Table A.17: Eight Contributing Experiments

#	Experiment	Key derivation result used in MATH-12
1	experiment math11 beta metric v0 run002	A_2 β decomposition: 90.4% cancellation, β^0 = 46.6%
2	experiment beta content a3 v0 run001	A_3 β decomposition: 99.5% cancellation, β^0 = 51.3%
3	experiment laporta pslq v0 run002	17/17 null: 6 independent, not polylogarithmic
4	experiment laporta a4 decomposition v0 run001	43 × Harvard, 48 ppb α shift
5	experiment laporta toroidal v0 run001	All β^0, 6/6 elliptic < 0.006%, ratio analysis
6	experiment laporta muon electron v0 run001	Sensitivity ratio 1.000, 2304% toroidal scaling, crossover 43 m_e
7	experiment remainder elliptic v0 run001	Control ratio 2.05, subtraction 6/6 improved 7-266 ×
8	experiment laporta attack3 v0 run002	Consistency check: k_{81} at 167 ppb, k_{83} at 25 ppm

All evidence from derivation chains. No PSLQ scans used as primary evidence. The PSLQ results (24/24 null, 17/17 null) serve only as independence proofs, not as structure discovery.

Table A.18: Predictions and Kill Switches

#	Prediction	Observable test	Kill condition
1	K(k)/ π appears in tokamak safety factor	Compare q tokamak to K(k plasma)/ π at measured k	No match for any physical k
2	QCD flux tube tension involves K(k QCD)	Lattice QCD string tension vs elliptic period	String tension purely rational or π -based
3	A_3 remainder's 1.8 ppm KE match reaches measured values	Derivation chain from K(0.99)E(0.99) to a e or α	Chain terminates without reaching measurement

#	Prediction	Observable test	Kill condition
4	Five-loop QED introduces new moduli	β decomposition of A_5 when analytically available	A_5 fully polylogarithmic (no new k values)
5	$k_{81} = 0.999994$ derivable from topology 81 propagator structure	Compute k from Feynman diagram masses and momenta	Computed k doesn't match extracted k
6	$k_{83} = 0.99713$ derivable from topology 83 propagator structure	Same	Same
7	Galaxy disk correction involves K(k gal)/ π	Hubble tension per-direction analysis with toroidal correction	No directional dependence, or correction doesn't match K form
8	The Lp generalization and the k generalization are orthogonal	β(p, k) = K p(k) for some generalized Lp elliptic integral	The two axes interact in an unexpected way

[**HOWL-MATH-12-2026**] β^0 Has Two Geometries. Github: https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-12-2026

[**HOWL-MATH-11-2026**] β = π/4. Github: https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-11-2026